Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

Relational Inventory Control & Stock Tracking System

SDG 9 – Industry, Innovation and Infrastructure

Presented By:

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INTRODUCTION

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- The Inventory Control and Stock Tracking System is a web-based application developed to manage product inventory efficiently.
- Many small businesses still rely on manual methods to track stock, which leads to errors and inefficiencies.
- This system automates stock management by allowing users to add products, update stock levels, and monitor transaction history.
- It ensures accurate stock tracking and prevents negative inventory issues. The system provides a dashboard for real-time data monitoring.
- It is developed using PHP and MySQL on XAMPP server. This project improves efficiency, accuracy, and data management

PROBLEM STATEMENT

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- Many small businesses manage inventory manually using notebooks or basic spreadsheets.
- This often leads to incorrect stock calculations and human errors.
- There is no proper tracking of stock inflow and outflow.
- Lack of transaction history makes monitoring and auditing difficult.
- Manual systems consume more time and reduce efficiency.
- Over-stocking or under-stocking can cause financial losses.
- Therefore, a reliable and automated inventory system is required.
- The system should manage stock accurately and maintain proper transaction records.

OBJECTIVE(S)

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- **To design and develop** a web-based Inventory Control and Stock Tracking System using PHP and MySQL.
- **To implement** stock IN and OUT management with transaction history and validation to prevent negative stock.
- **To prevent** negative stock and ensure data integrity.
- **To provide** a real-time dashboard for monitoring total products, stock, and transactions.
- **To improve** operational efficiency and reduce manual errors in inventory management.

PROPOSED SYSTEM

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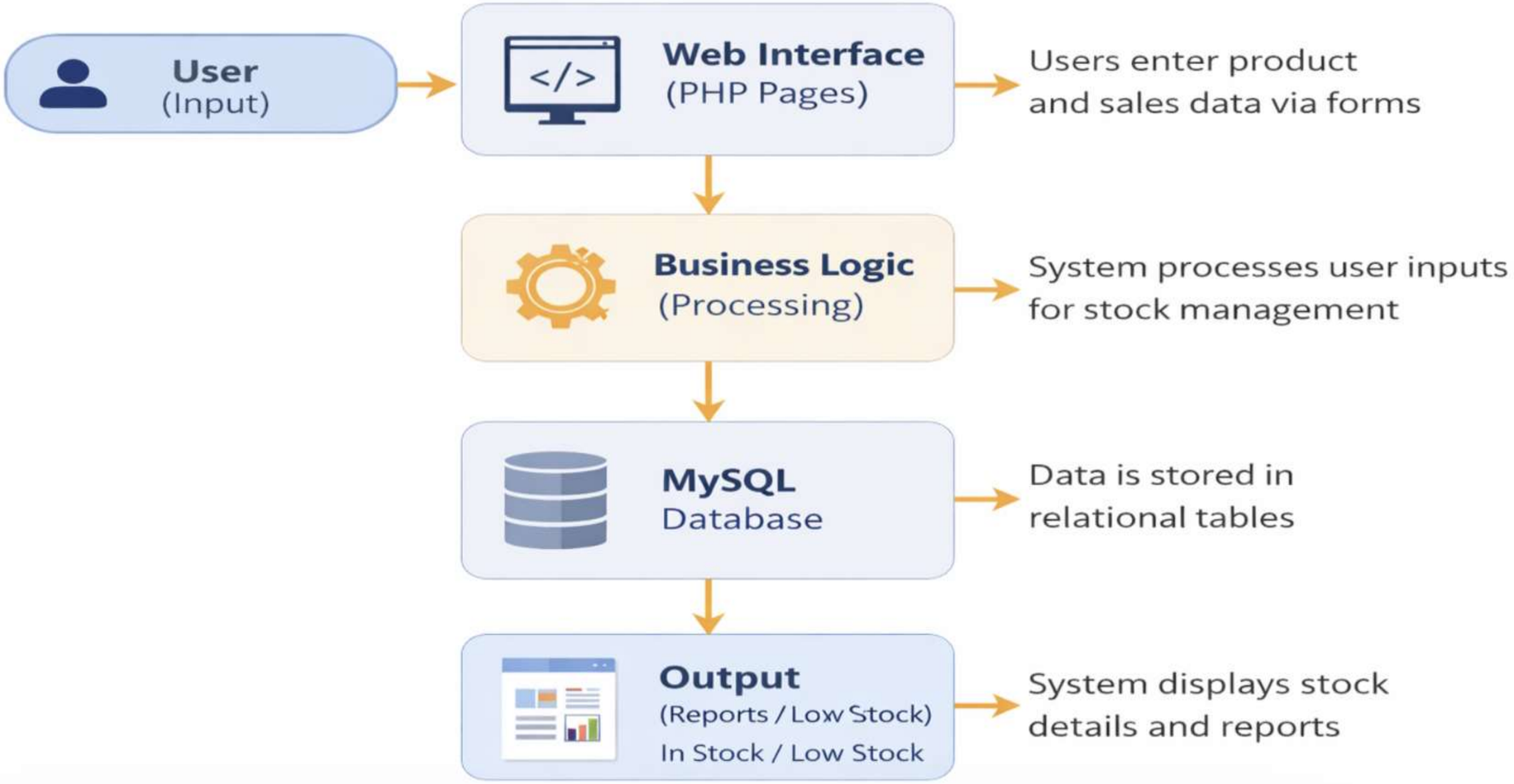
- The proposed system is a web-based Inventory Control and Stock Tracking application developed using php and MySQL.
- It allows users to add, delete, and manage products efficiently.
- The system enables stock IN and OUT operations with automatic quantity updates.
- Every stock update is recorded in a transaction table for tracking and auditing.
- The system prevents negative stock by validating quantity before processing transactions.
- A real-time dashboard displays total products, stock levels, and transaction count.
- Search functionality allows quick access to transaction history.
- Compared to manual systems, it improves accuracy, reduces human errors, and enhances operational efficiency.

PROPOSED SYSTEM ARCHITECTURE



PROPOSED SYSTEM ARCHITECTURE

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- **Input**

  User enters product details, stock updates (IN/OUT), and search queries through web forms.

- **Processing**

  PHP processes the user request, validates stock quantity, performs calculations, and executes SQL queries.

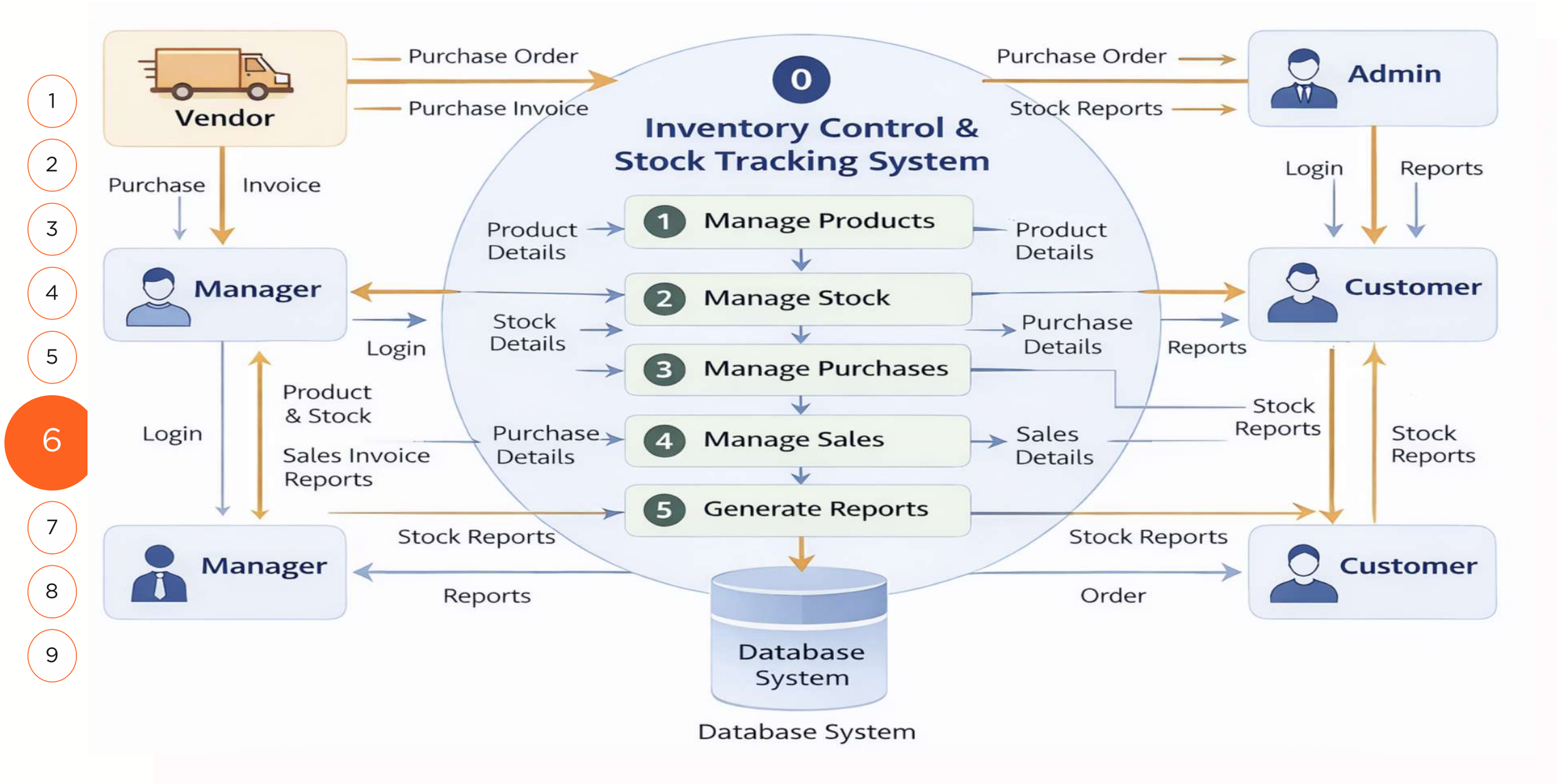
- **Database**

  MySQL database stores product details and transaction records using relational tables.

- **Output**

  The system displays updated stock information, transaction history, and dashboard summaries in the browser.

WORKFLOW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Vendor
Purchase Order
Purchase Invoice
Purchase
Invoice
Manager
Login
Product & Stock
Sales Invoice Reports
Login
Manager
Stock Reports
Reports
0
Inventory Control & Stock Tracking System
Product Details
1 Manage Products
Product Details
2 Manage Stock
Stock Details
Purchase Details
3 Manage Purchases
Purchase Details
4 Manage Sales
Sales Details
5 Generate Reports
Database System
Database System
Purchase Order
Stock Reports
Admin
Login
Reports
Customer
Reports
Stock Reports
Stock Reports
Customer
Stock Reports
Order

TECHNOLOGIES USED

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- **Frontend**: HTML5, CSS3
- **Backend**: PHP
- **Database**: MySQL
- **Server**: XAMPP (Apache Server)
- **Tools**: VS Code ,phpMyAdmin
- **Hardware/Software**: Windows OS, XAMPP Server

MODULE DESCRIPTION

Module 1: Product Management

Handles adding and deleting products from the system.

Module 2: Stock Management

Manages stock IN and OUT operations with validation.

Module 3: Transaction Management

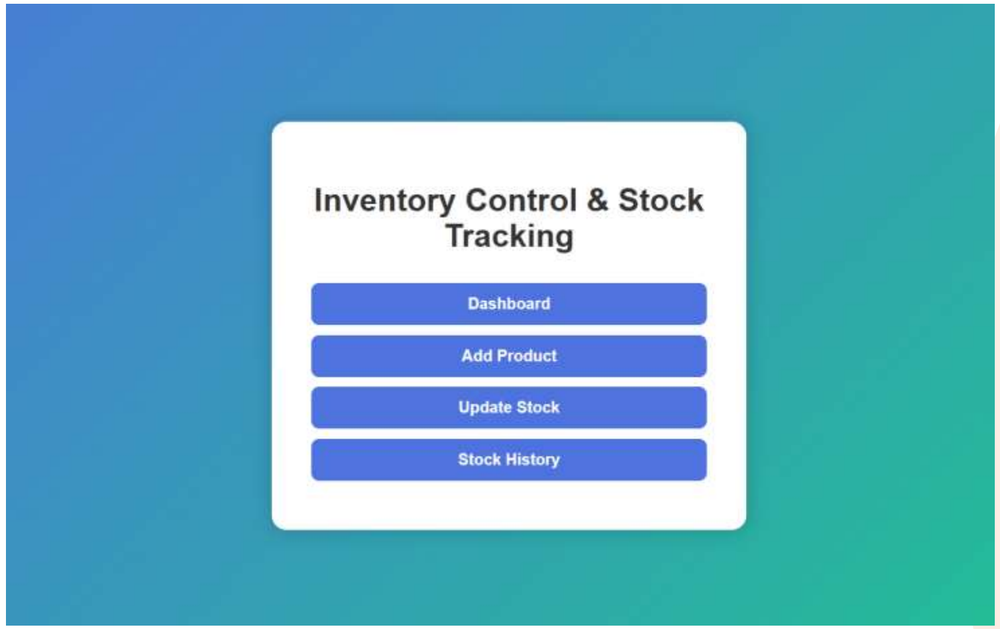
Records and displays stock transaction history using INNER JOIN.

Module 4: Dashboard Module

Displays total products, total stock, and total transactions summary.

IMPLEMENTATION SCREENSHOTS

- The Home Page serves as the main navigation interface of the Inventory Control & Stock Tracking System.
- It provides quick access to all major modules of the system.
- Users can easily navigate to Dashboard, Add Product, Update Stock, and Stock History sections.
- The interface is designed to be simple, user-friendly, and responsive.
- It ensures smooth navigation and better user experience.



IMPLEMENTATION SCREENSHOTS

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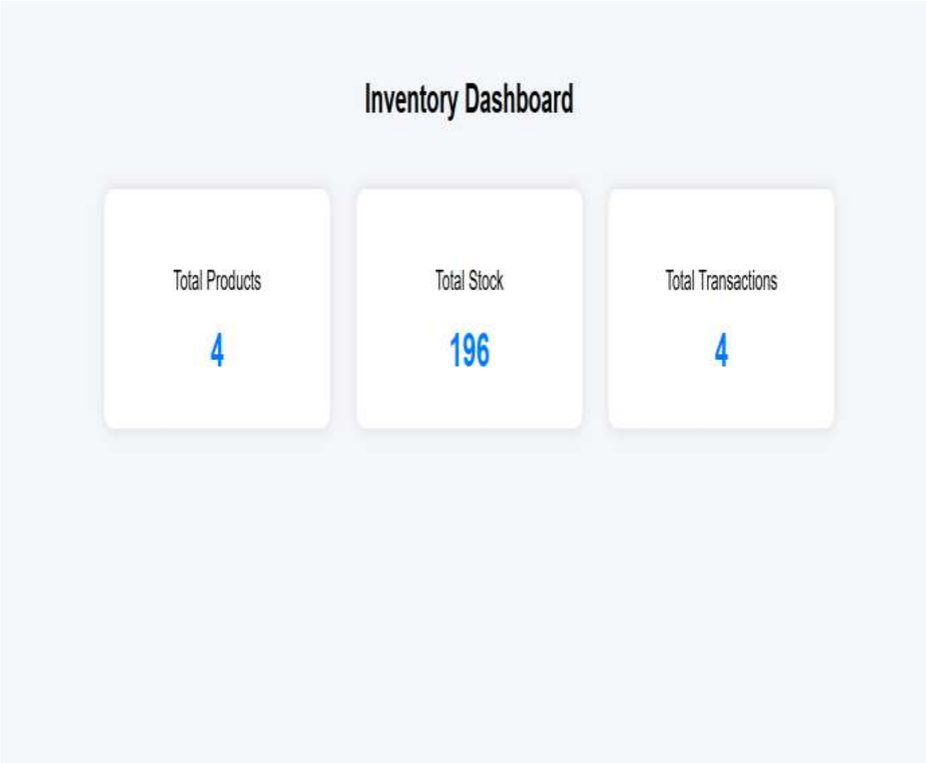

DASHBOARD MODULE

- The Dashboard provides a summary view of the inventory system.
- It displays important statistics such as Total Products, Total Stock, and Total Transactions.
- These values are dynamically fetched from the MySQL database.
- It helps users quickly understand the current inventory status.
- The dashboard improves decision-making by showing real-time data.
- It reduces the need to manually calculate stock information.
- The interface is simple, clean, and easy to interpret.
- It acts as the central monitoring panel of the system.

IMPLEMENTATION SCREENSHOTS

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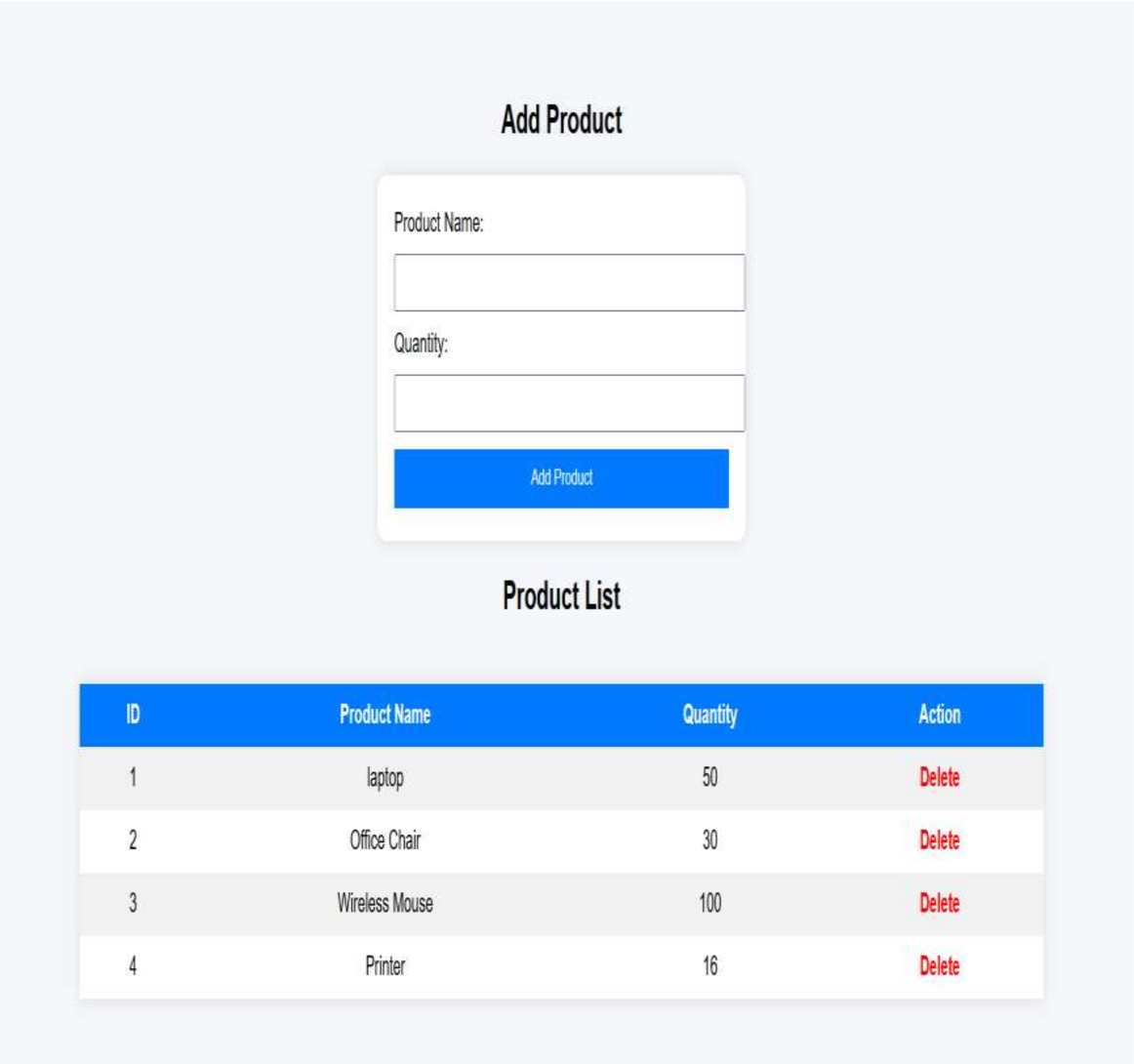

ADD PRODUCT MODULE

- The Add Product module allows the admin to add new products into the inventory system.
- The user enters the product name and quantity through a form.
- The data is sent to the PHP backend for processing.
- PHP validates the input and inserts the record into the MySQL database.
- The product list is automatically updated after insertion.Each product is assigned a unique ID.

IMPLEMENTATION SCREENSHOTS

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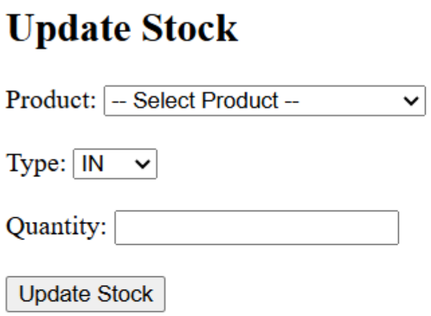

UPDATE STOCK MODULE

- The Update Stock module allows the admin to modify existing product quantities.
- The user selects a product from the dropdown list.
- The admin chooses the transaction type as IN (stock added) or OUT (stock removed).
- The quantity field is used to enter the stock amount.
- PHP processes the request and updates the stock value in the MySQL database.

IMPLEMENTATION SCREENSHOTS

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STOCK HISTORY

Stock Transaction History

Search by product name | Search

ID	Product	Type	Quantity	Date
4	Printer	IN	8	2026-02-22 14:53:53
3	Wireless Mouse	IN	50	2026-02-22 14:53:47
2	Office Chair	IN	15	2026-02-22 14:53:35
1	laptop	IN	25	2026-02-22 14:53:27

ADVANTAGES & LIMITATIONS

- **Advantages**
  - Reduces manual errors
  - Prevents negative stock
  - Easy to use interface
  - Real-time stock updates
  - Maintains transaction history

- **Limitations**
  - Designed for small-scale businesses
  - No multi-user login system
  - Requires local server (XAMPP)

Thank you